Department of Physics
University of California, San Diego
Physics 1A, Summer Session 1, 2026

Instructors:
Jiashu Han, jih178@ucsd.edu
Office Hours[1]: I scheduled office hours right after lecture in the classroom to make it accessible to students.
Monday-Thursday 10:50-11:50am in MOS 0114 (right after lecture)
Monday and Thursday 2:00-2:50pm in the lounge outside Mayer 4238
I added additional options for office hours to accommodate students who are unable to attend the after-class office hours

Teaching Assistant:
Michael Skripalsh, mskripalsh@ucsd.edu
Office Hours: The TA inadvertently scheduled OH at the same time as mine, but this should in general be avoided.
Monday 2:00-3:00pm, Mayer 5712 and on zoom
Friday 11:00am-12:00pm, Mayer 2218

Course Webpage: Login through http://canvas.ucsd.edu.
All assignments will be due through Kudu or Gradescope (both can be accessed through Canvas).

Required Textbook: Kudu Online Textbook "Physics for Life Sciences" **Course ID: bd9n5l**

I wanted to start the first page with a clear description of what students are expected to do - when and where everything happens.

Course Meeting Times, A00[2]:

Lecture (A00)	MTuWTh	9:30-10:50am	MOS 0114
Problem Solving (optional)	Friday	10:00-11:00am	Mayer 2218
Final Exam	Friday, Jul 31	8:00-11:00am	MOS 0114

What will be done for each type of session:

- ***Lectures***: This is our main meeting time. We will go through the main equations and ideas, leaving ample time for examples/interactive questions to review concepts and problem-solving strategies.
- ***Quizzes***: There will be 4 quizzes during weeks 2, 3, 4, and 5 to help you track your progress and identify areas you might need to put more work into. The quizzes will take place during the Tuesday lecture time. Make sure you can attend each of these quizzes! An equation sheet will be provided, but you will need to bring a calculator that can't connect to the internet. No scantron or blue book is required - you'll write directly on the exam.

This allows students to ask questions on Monday.

Made available a week in advance on Canvas so students can use it as a guide to study

- ***Problem Solving Session*** (optional): There will be an optional weekly problem solving session where the TA will lead students through one or two example problems.
- ***Final Exam***: This is how we assess how much you have learned from this course. The final exam will have the same format as the quizzes, but longer in length and comprehensive in topics covered.

[1] This is just a place for us to talk face to face - about anything, even if you have no questions at all!
[2] All of the following due dates/times are San Diego time (PDT).

Clarifies what office hours are since many first-year students are unfamiliar with the idea and would under-utilize OH as a result.

The second page thoroughly explains what students are expected to get out of this course.

About this course: Physics 1A is the first of a three-quarter introductory physics sequence intended for life-science majors. In this quarter, you will learn about: Equilibrium and motion of particles in one and two dimensions in the framework of Newtonian mechanics, force laws (including gravity), energy, momentum, rotational motion, conservation laws, and fluids. We will explore how all of these concepts fit into our daily and professional life, and also prepare you to succeed in the next chapters of this course sequence!

Prerequisites: MATH 10A or 20A. Recommended: MATH 10B or 20B.

Learning Outcomes

What will you learn in this course? Upon the successful completion of Physics 1A, you will be able to

Specific to course material

- Analyze simple problems using *units* and *dimensions* and make estimations about physical quantities using *scaling* arguments.
- Apply the equations of *kinematics* to solve problems involving moving objects with constant acceleration in one or two dimensions.
- Analyze different types of forces on objects using *force diagrams* and predict the motion and behavior of these objects using *Newton's Laws*.
- Describe the properties and behavior of *fluids*.
- Identify conserved quantities such as *energy* and *momentum* in a problem and use conservation laws to simplify the problem.
- Analyze and solve problems involving *rotating objects*.

More transferrable skills

More generally, you will be able to

- Identify quantities of motion in a described scenario (even when they are not explicitly mentioned by name)
- Recognize the physical processes in a described scenario
- Identify the goal of a problem and build the solution to the problem around the goal
- Break down problems into stages and components based on the identified physical processes and quantities
- Use scaling arguments to quickly predict the change in a system's behavior without fully solving the problem
- Distinguish between relevant and irrelevant information when solving a problem

This part covers how students are assessed in the course:

How you will be graded: Your final course grade will be determined according to the following:

Helps students reinforce what they just learned

- **Daily Review** (15%): 3-4 problems are assigned on Canvas after each lecture for you to review the content covered in lecture. They are due at the end of the same day.

The daily review assignments are in general well received by the students, but they place constraints on what I need to cover during the lectures.

This reflects the quizzes' role as formative rather than summative assessments. It worked as intended and I received feedback from students that it relieved a lot of their stress and allowed them to learn from their mistakes.

- **Weekly Homework** (15%): These are weekly problem sets you turn in through Kudu and Gradescope (more on this below). You will also be given the opportunity to correct your previous quiz, which can refund you up to 25% of the lost quiz points. We will drop 1 lowest graded assignment for any reason (including emergency).
- **Quizzes** (30%): The lowest score out of the 4 quizzes will be dropped. **Note**: This is our emergency drop policy — if you need to miss a quiz due to some emergency, this policy is designed to address that. There will be **no** makeup quizzes.

The emergency drop policy reduces logistical issues and also lowers the stakes for the quizzes, which are intended as formative assessments,

- **Final Exam** (40%): You must take the final exam at the scheduled time and location. If you are unable to do so, please consider dropping the course.

This incentivizes participation and offers the students oppotunities to think actively in class.

- **Participation** (extra credit): The class participation grade is based on poll questions during lectures that you will submit via Google Forms. You only need to answer 80% of all questions to receive full credit. It can replace 10% of your final exam if it helps your overall grade. There will be no makeup opportunities for class participation.

Taking all quizzes is an important aspect of this course. If there is some emergency and you cannot attend a quiz, the above policy is designed to automatically take this into account, and you won't need to contact the instructor or TA for this. If you have an extended hospital stay (1+ weeks), or other long-term issue that you think the instructor should know about, please email the instructor as soon as the situation arises to meet to discuss your particular circumstances.

The tentative grade cutoffs for this course will be as follows:

A	A-	B+	B	B-	C+	C	C-	D
$> 93\%$	$> 89\%$	$> 85\%$	$> 81\%$	$> 78\%$	$> 74\%$	$> 70\%$	$> 66\%$	$> 55\%$

If you are taking the class P/NP, you need at least the equivalent of a "C-" to pass.

I intentionally included the most essential information in the first two and a half pages (everything above). Everything below is intended to add clarity or provide additional context. I made this choice because it is unlikely that most students would read through a six-page document in full. This following section explains the assignments in more detail.

Daily Review: The daily post-lecture assignments will be assigned as quizzes on Canvas and due by the end of the day. The point of these assignments is to make sure you're keeping up with the material, which is especially important given the fast pace of the course. Going over the basic material at home will make the (very quick!) lectures more comprehensible.

You may submit as many times as you want before the deadline, but only the last submission will be graded.

This setup permits students to learn through trial and error, but prevents them from just randomly guessing.

Weekly Homework: There will be a weekly problem set due towards the end of each week based on the material covered during the preceding lectures. Each problem set consists of: (1) 20 multiple choice questions that you submit directly on the Kudu textbook, which are automatically graded; and (2) fully written solutions to 10 (picked by us) of the 20 Kudu problems submitted on Gradescope. For each Gradescope assignment, we will randomly pick 1-2 problems to grade on

The Gradescope portion encourages students to fully attempt the problems and solve them with the proper techniques and strategies, rather than through guesswork. I received mixed feedback - some students appreciate the practice they got from writing out the full solutions, while others commented on the confusing nature of repeating the problems on a different platform.

the correctness of your written solutions[3].

If you have questions about any of the HW problems, we are happy to answer them on Piazza or during the instructor or TA office hours! Sometimes you need more time to work on the problems or get help from us, and that's ok! Unless specified otherwise, you may still submit your written work through Gradescope up to 72 hours (3 days) beyond the usual deadlines listed on the course schedule.

The Kudu portion of each weekly homework assignment is worth 20 points and purely graded on accuracy. The grade for the Gradescope portion consists of the following parts:

- *Accuracy* (10 points): This part of the homework is graded based on the correctness of your response. Your written work grade for an entire assignment will be based on your responses for 1 to 2 randomly chosen problems (out of 10).
- *Organization* (5 points): Your written work must be neat, legible, and organized. When you turn in your written work on Gradescope, it will ask that you select the pages of your work that correspond to each problem. Make sure you select the correct pages - doing so will automatically earn you these points. If the TA/grader has to go through your homework because you did not select the appropriate pages, you will not receive the organization points for your assignment.

Making timeliness an organic part of the grade, instead of presenting it as a "late penalty." It did not result in a significant number of late submissions.

- *Timeliness* (5 points): You will receive full points for turning in before the specified deadline. You will receive 1 out of 5 points for turning in before the solutions are posted 24 hours after the deadline. You will receive no timeliness points for turning in after the solutions are posted, and we will not accept any request for extension beyond this time. For the Gradescope assignments, you will not be able to submit on Gradescope or receive any credit for the assignment beyond 72 hours after the deadline.

This section also provides a rationale for why students should submit on time.

Solutions to the Kudu problem set will be available on Kudu exactly 24 hours after the deadline. It is your responsibility to "revise" / "correct" your homework assignment. Most of the time, students can compare their work to the solutions on their own and understand whether their work was okay or not; if this is not the case for you, please come to office hours (instructor or TA) for clarification.

Topics Covered on Quizzes/Exams:

The quizzes and the final exam are carefully designed to target the course materials referenced here. This would help students in their preparation for these assessments. Some SET reviews I received noted that the quizzes covered more than "just the previous week's material" which implied that this piece of information never reached these reponding students - Next time I would probably move it up to a more obvious location (the first couple of pages) and actively remind them in class.

1. **Quiz 1** will cover topics from Lectures 1-4, **DR 1-4** and **Weekly HW1**
2. **Quiz 2** will cover topics from Lectures 1-7, **DR 1-7** and **Weekly HW1-2**
3. **Quiz 3** will cover topics from Lectures 1-10, **DR 1-10** and **Weekly HW1-3**
4. **Quiz 4** will cover topics from Lectures 1-13, **DR 1-13** and **Weekly HW1-4**
5. **Final exam** will cover Lectures 1-16, **DR 1-15** and **Weekly HW1-5**

[3]Because there are ∼100 students in the course, we will not have the resources to review every problem when we determine your grade for HW assignments.

Discussion Board: We will use **Piazza** — an online course discussion board where you can ask / answer questions about the course material (including homework). If you send me an email about a specific problem, I will generally redirect you to Piazza. There are three clear benefits:

1. Everyone can see one another's questions/responses, which helps multiple people have the same question.
2. The TA and I can use LaTeX to make equations look nice/understandable.
3. Students can help one another learn the material. Additionally, you may receive extra credit (0.5% of the course grade) for posting more than 4 responses that are endorsed by the instructors. Small incentive for participation on Piazza. It was effective for those who used Piazza and they made many meaningful contributions, but overall only around a quarter to a third of the class used Piazza.

The link for the Piazza page is on Canvas.

Academic Integrity: Please read the College Policies section of the UCSD's Policy on Integrity of Scholarship. These rules will be enforced. Cheating includes, but is not limited to: submitting another person's work as your own (or allowing someone to do so, e.g. having your classmate answer poll questions for you), copying from any person/source, and using any unauthorized materials or aids (whether at home or during quizzes/exams).

We encourage collaboration or the use of online resources (including ChatGPT and similar AI tools) to help you enhance your learning process, as long as the work you do and submit is your own. By the time you start writing your own solution to turn in, you should not be looking at any other source. You should know the rough outline of the solution well enough that you don't need to reference something line-by-line. For homework assignments, the following practices are considered cheating:

- Directly copying from an online solution, a peer's solution, a Chegg solution, an AI-generated (ChatGPT or similar) solution, or shared work (on discord, for example)
- Plagiarizing a solution but changing variable names
- Soliciting help online via Chegg, Quora, etc.

If suspected, you might be asked to rework similar problems in a Zoom one-on-one meeting with the instructor.

All course material (problems sets, solutions, worksheets, videos, etc.) are for the private use of registered students only. No materials should be distributed (or links shared) in physical or electronic form to others without prior consent of the instructor.

Questions on what constitutes an academic integrity violation should be addressed to the instructors; any violation will result in immediate reporting to the UC San Diego Office of Academic Integrity and can result in an automatic "F" for the course at the discretion of the instructor.

Education Research: Data from this class may be used (anonymously!) to determine the effectiveness of the pedagogical techniques used in this course. For more information / to opt-out, please email the instructor.

These deadlines are also included in the lecture slides as reminders.

Add/Drop Deadlines: Add/drop deadlines can be found here. The deadline to drop without a "W" is Jul 10 (Fri Week 2). The deadline to change grading option, change units, or drop with a "W" is Jul 17 (Fri Week 3).

Satisfactory Academic Progress (SAP) Satisfactory Academic Progress (SAP) refers to the academic standards students must maintain to remain eligible for federal, state, and institutional financial aid. If you are receiving financial aid, please ensure you review the SAP requirements and the appeals process.

A detailed plan for the course. It is fully integrated into the course Canvas page and allows students to navigate to everything they need in the course from a single calendar that's intuitive to follow. I received very positive feedback on the ease of use of the Canvas site and the course schedule.

Detailed Course Schedule (Tentative)

Week/Day	*Topics/Lectures/Assignments*	*Sections / Chapters in in Kudu textbook to read before lecture*
WEEK 1:	**Introduction, Kinematics**	
Jun 29 (M)	Lecture 1: Introduction	Ch. 1, 2
Jun 30 (Tu)	Lecture 2: 1D Kinematics, Free Fall	Ch. 3
Jul 1 (W)	Lecture 3: Vectors, 2D Kinematics	Ch. 4
Jul 2 (Th)	Lecture 4: Projectile Motion, Forces	Ch. 4, 5
	Weekly HW1 due Saturday, July 4 11:59pm	
WEEK 2:	**Forces, Newton's Laws**	
Jul 6 (M)	Lecture 5: Forces, Newton's Laws	Ch. 5
Jul 7 (Tu)	**Quiz 1**	
Jul 8 (W)	Lecture 6: Force Diagram, Gravity, Tension	Ch. 5, 6
Jul 9 (Th)	Lecture 7: Friction, Springs, Circular Motion	Ch. 6, 7
	Weekly HW2 due Friday, July 10 11:59pm	
WEEK 3:	**Fluids, Energy**	
Jul 13 (M)	Lecture 8: Pressure, Buoyancy, Fluid Flow, Drag	Ch. 8, 9
Jul 14 (Tu)	**Quiz 2**	
Jul 15 (W)	Lecture 9: Energy, Work	Ch. 10
Jul 16 (Th)	Lecture 10: Power, Dissipative Forces, Momentum	Ch. 10, 11
	Weekly HW3 due Thursday, Jul 16 11:59pm	
WEEK 4:	**Conservation Laws**	
Jul 20 (M)	Lecture 11: Conservation of Momentum	Ch. 11
Jul 21 (Tu)	**Quiz 3**	
Jul 22 (W)	Lecture 12: Collisions	Ch. 11
Jul 23 (Th)	Lecture 13: Rotational Kinematics	Ch. 12
	Weekly HW4 due Thursday, Jul 23 11:59pm	
WEEK 5:	**Rotational Motion, Review**	
Jul 27 (M)	Lecture 14: Torque, Moment of Inertia	Ch. 12
Jul 28 (Tu)	**Quiz 4** In practice, this quiz happened too close to the final exam, and it made the last week very stressful for everyone. I would consider reducing the total number of quizzes to 3.	
Jul 29 (W)	Lecture 15: Rotational Energy, Angular Momentum	Ch. 12, 13
	Weekly HW5 due Wednesday, Jul 29 11:59pm	
Jul 30 (Th)	Lecture 16: Review I did not plan for any new material for the last lecture as it was less than a day before the final exam and that was not enough time for the students to digest new information.	
Jul 31 (F) 8:00am	**FINAL EXAM**	(all of the above)